Figure 1c: Pearson Correlation Coefficients with HOMA-IR

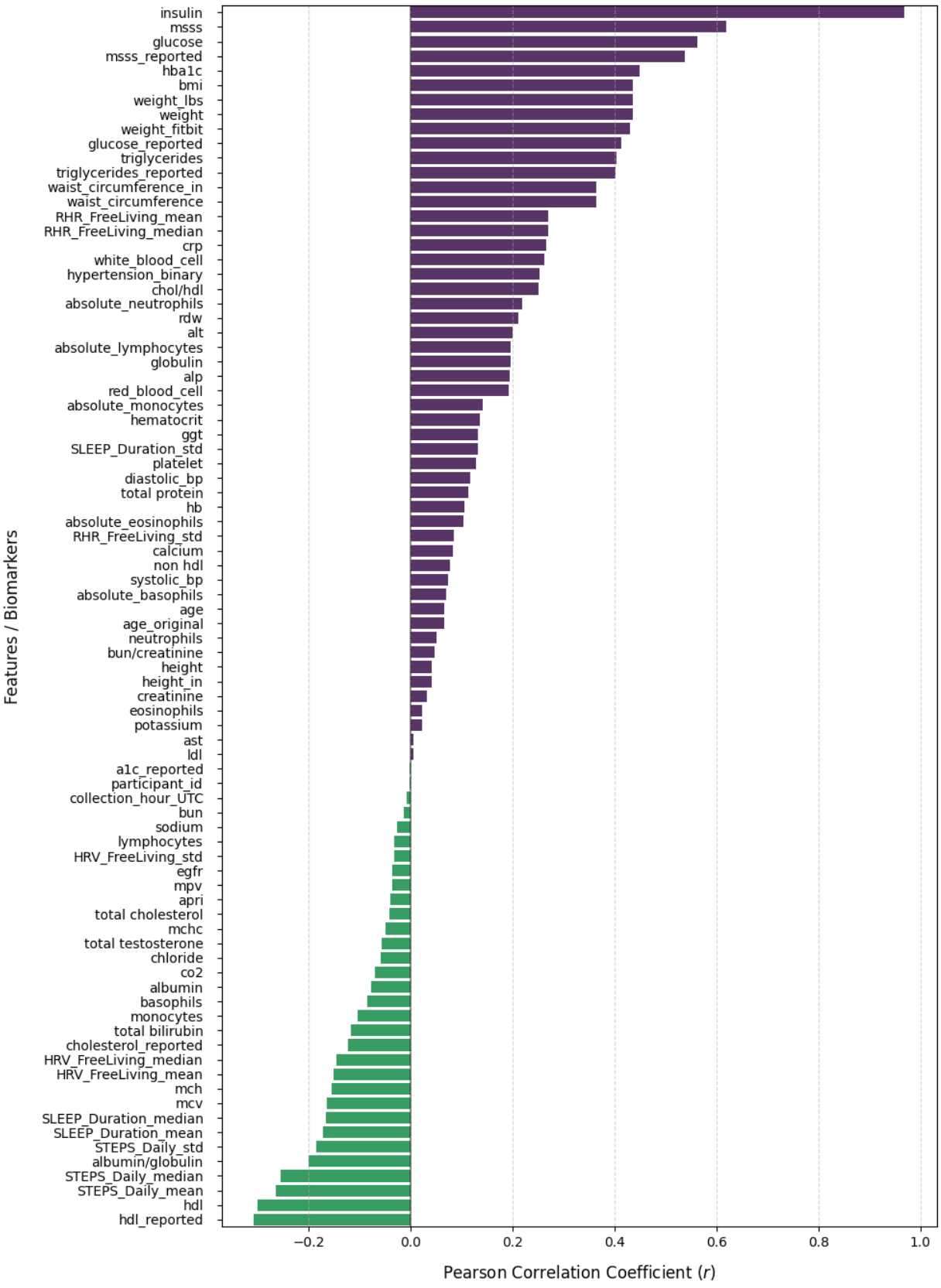
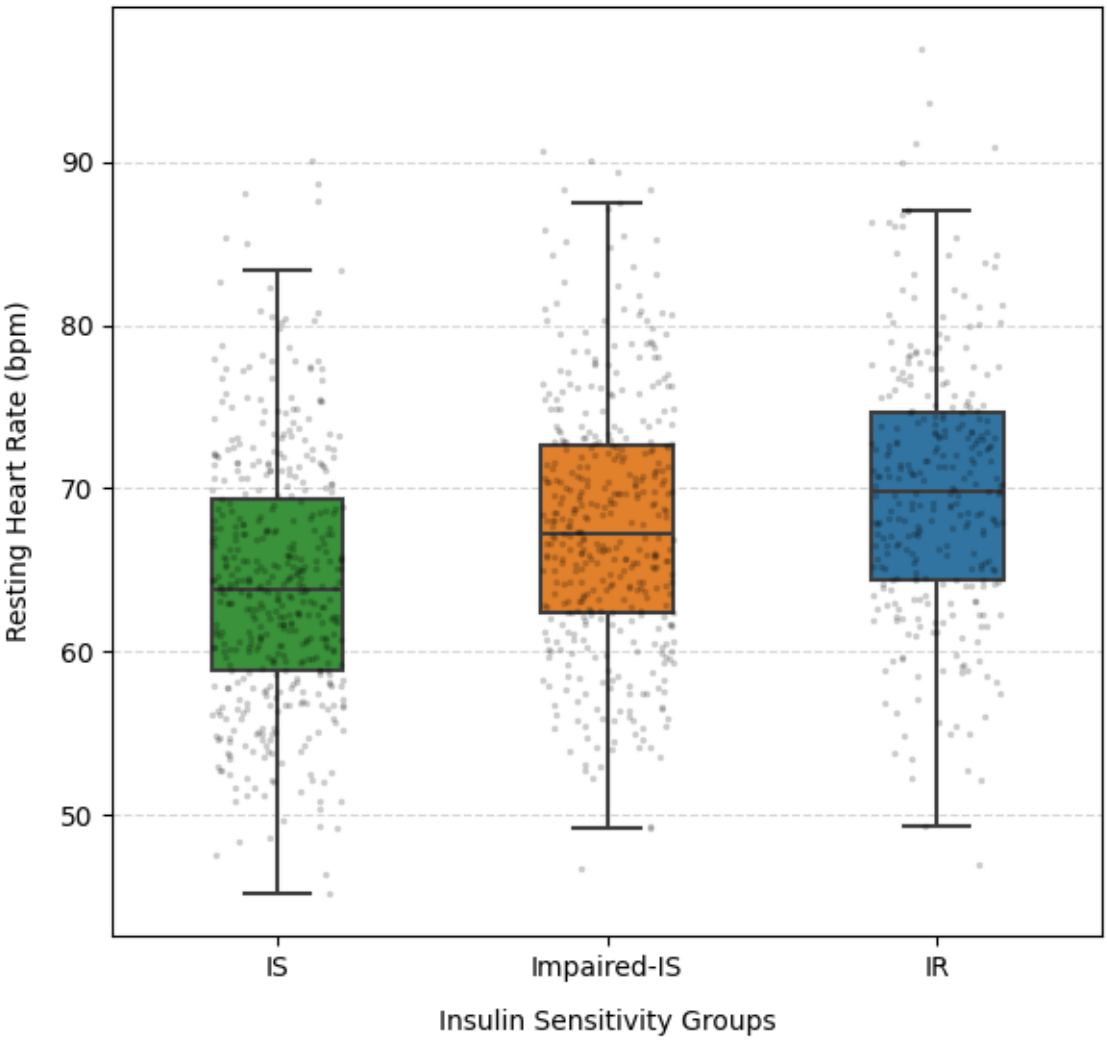


Figure 1d: Resting Heart Rate (RHR)



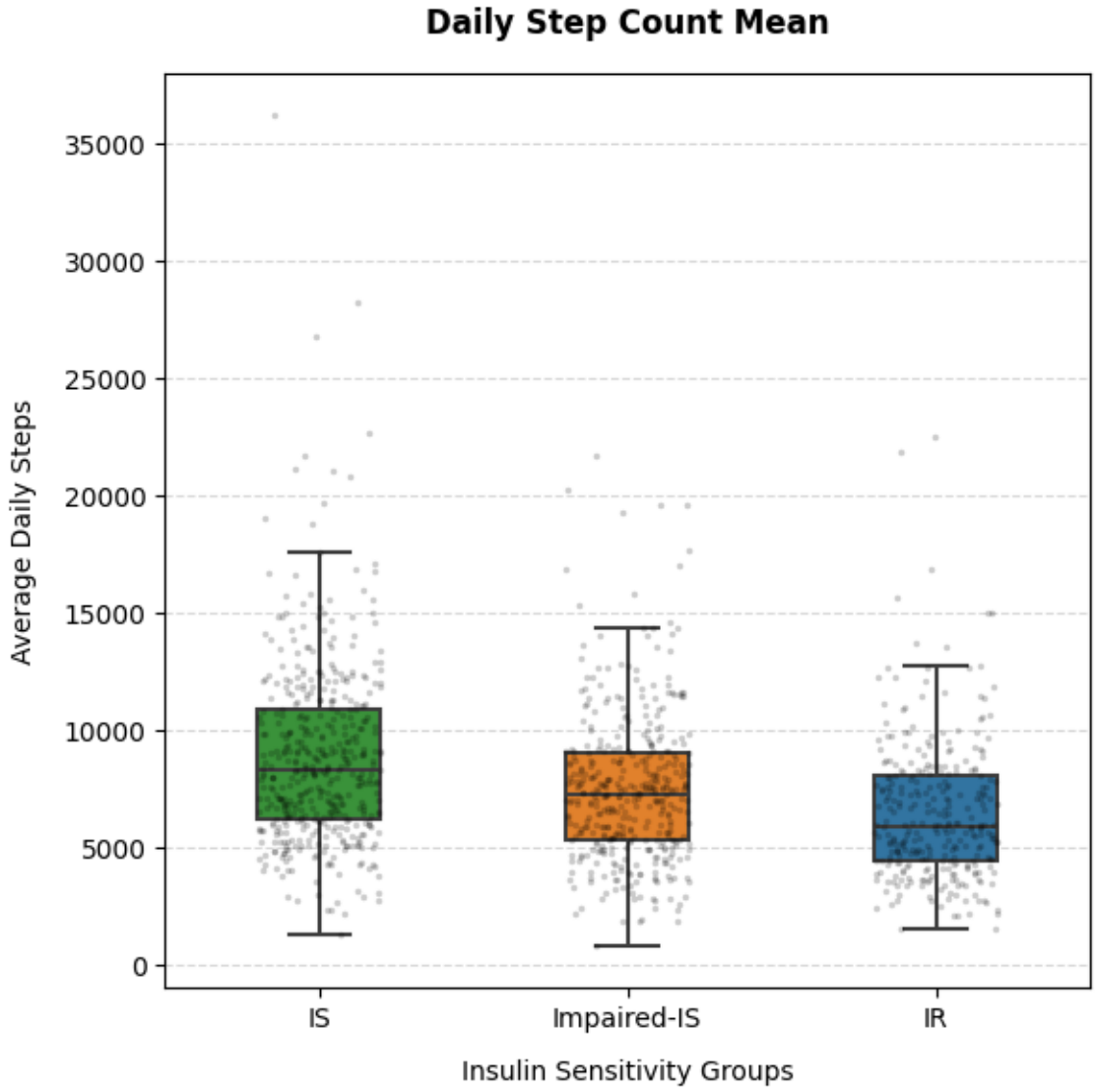

Daily Step Count Mean
35000
30000
25000
20000
15000
10000
5000
0
Average Daily Steps
IS
Impaired-IS
IR
Insulin Sensitivity Groups

Figure 1f: HRV (RMSSD)

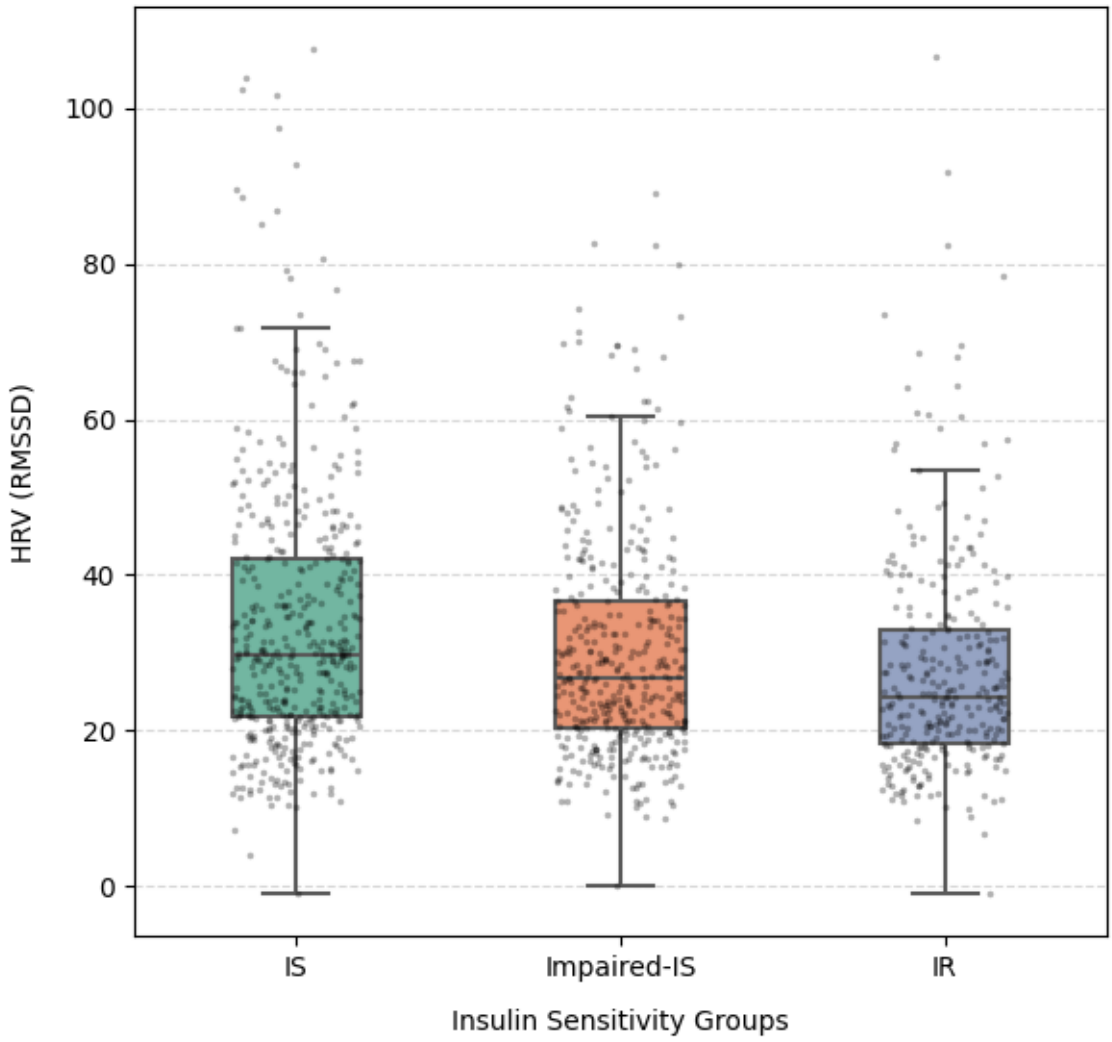


Figure 1g: Triglycerides across Insulin Sensitivity Groups

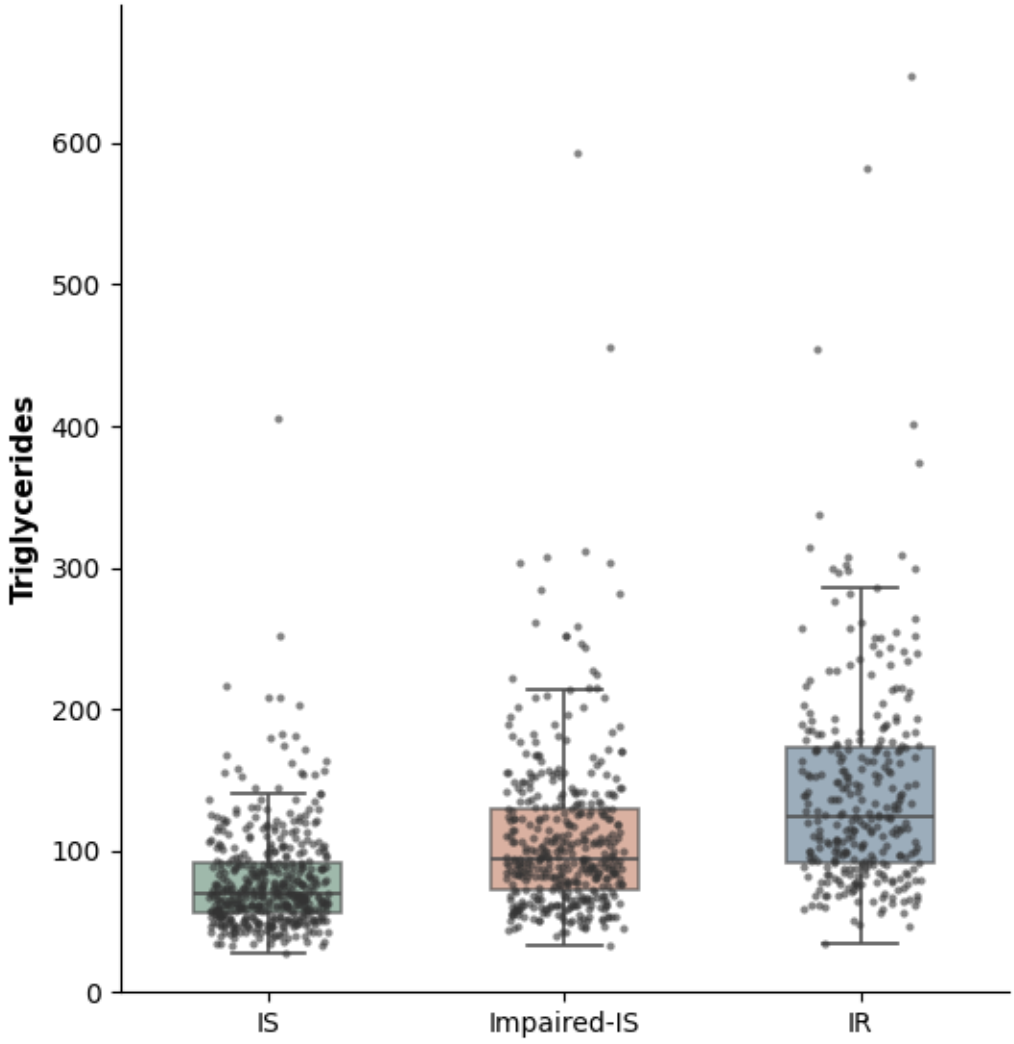


Figure 1h: HDL Cholesterol across Insulin Sensitivity Groups

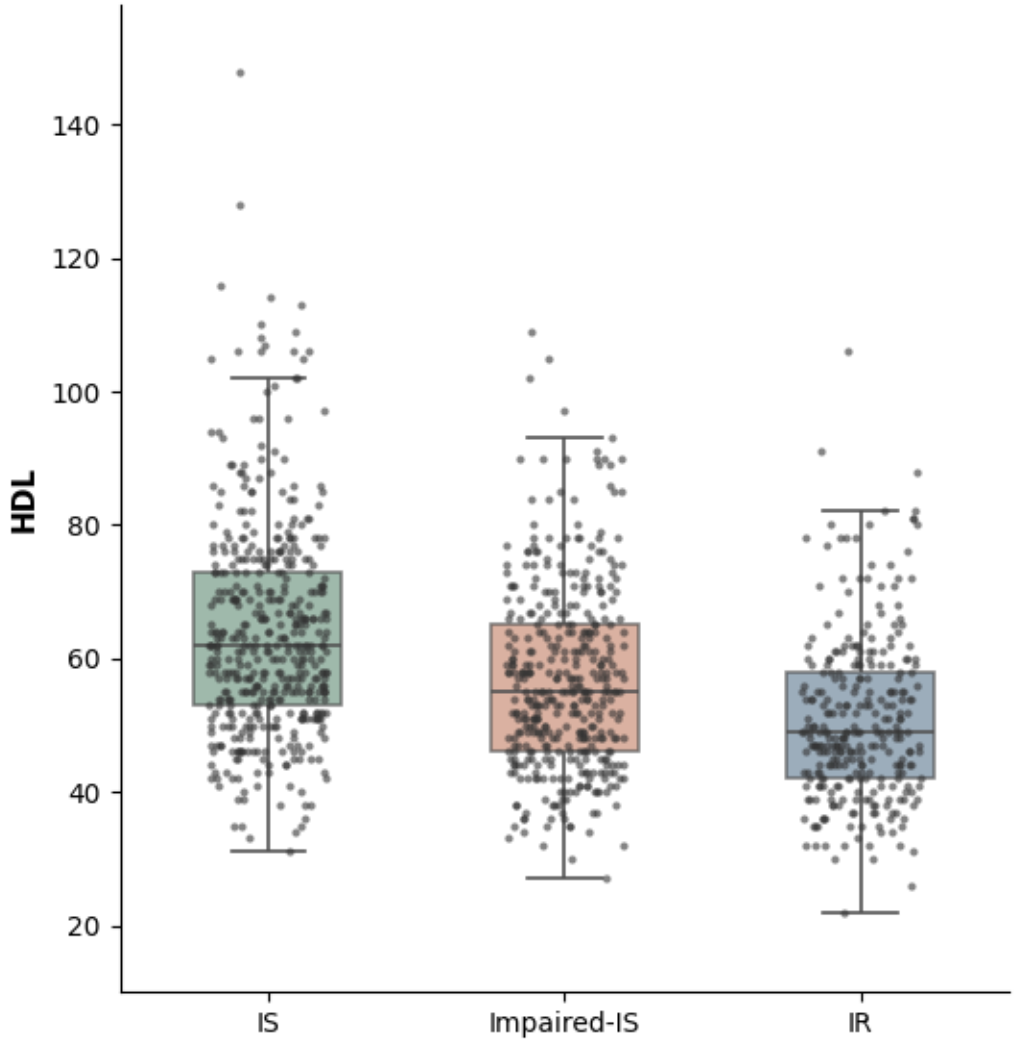
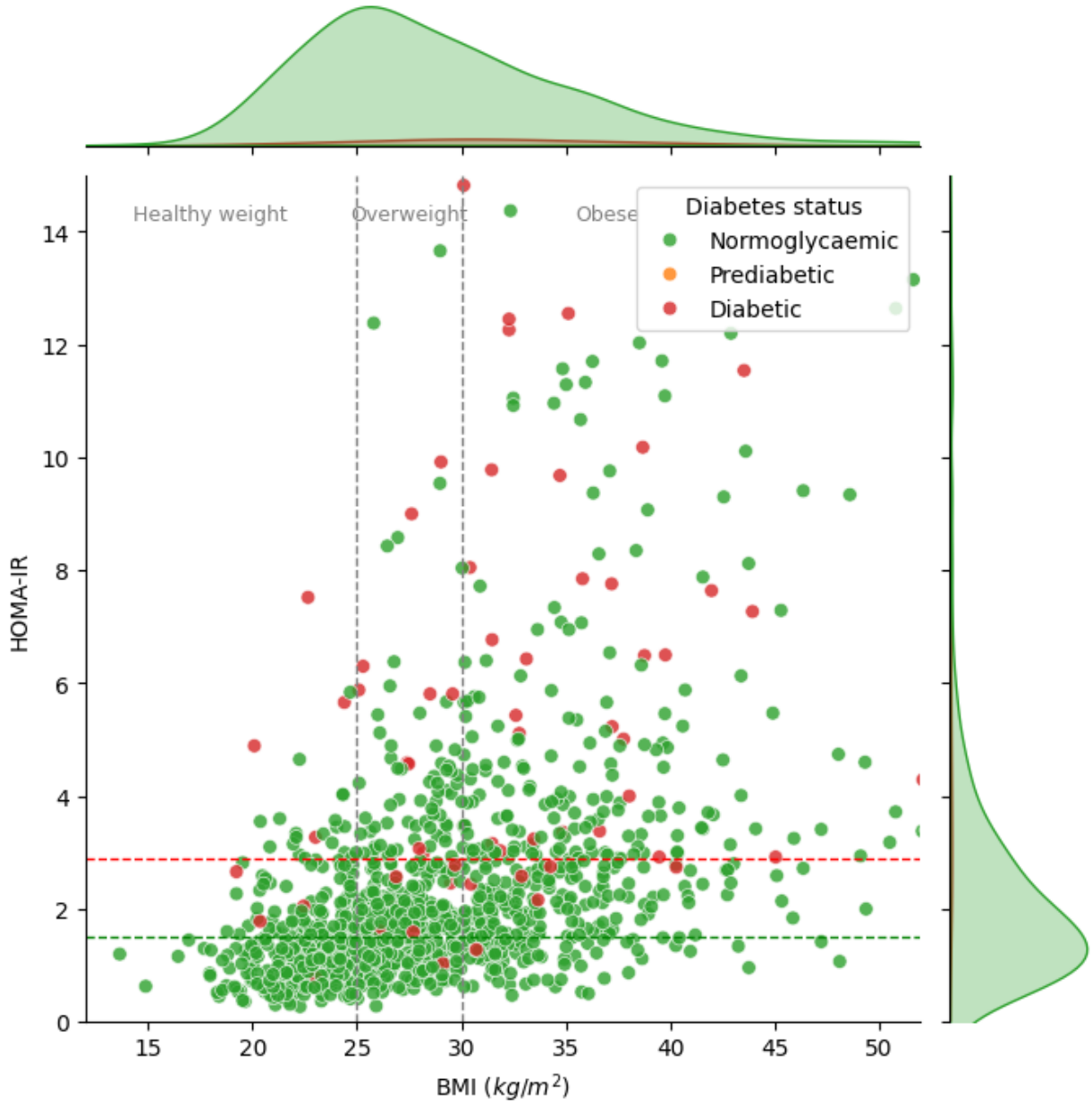


Figure 1j: Joint Scatter Plot with Marginal Distributions



ملاحظة: لا يوجد في الملف بيانات لتحديد ال

prediabetic

لذا الرسم يظهر بلونين فقط، الاحمر والاخضر

Figure 1k: Distribution of Participants (Paper Methodology)

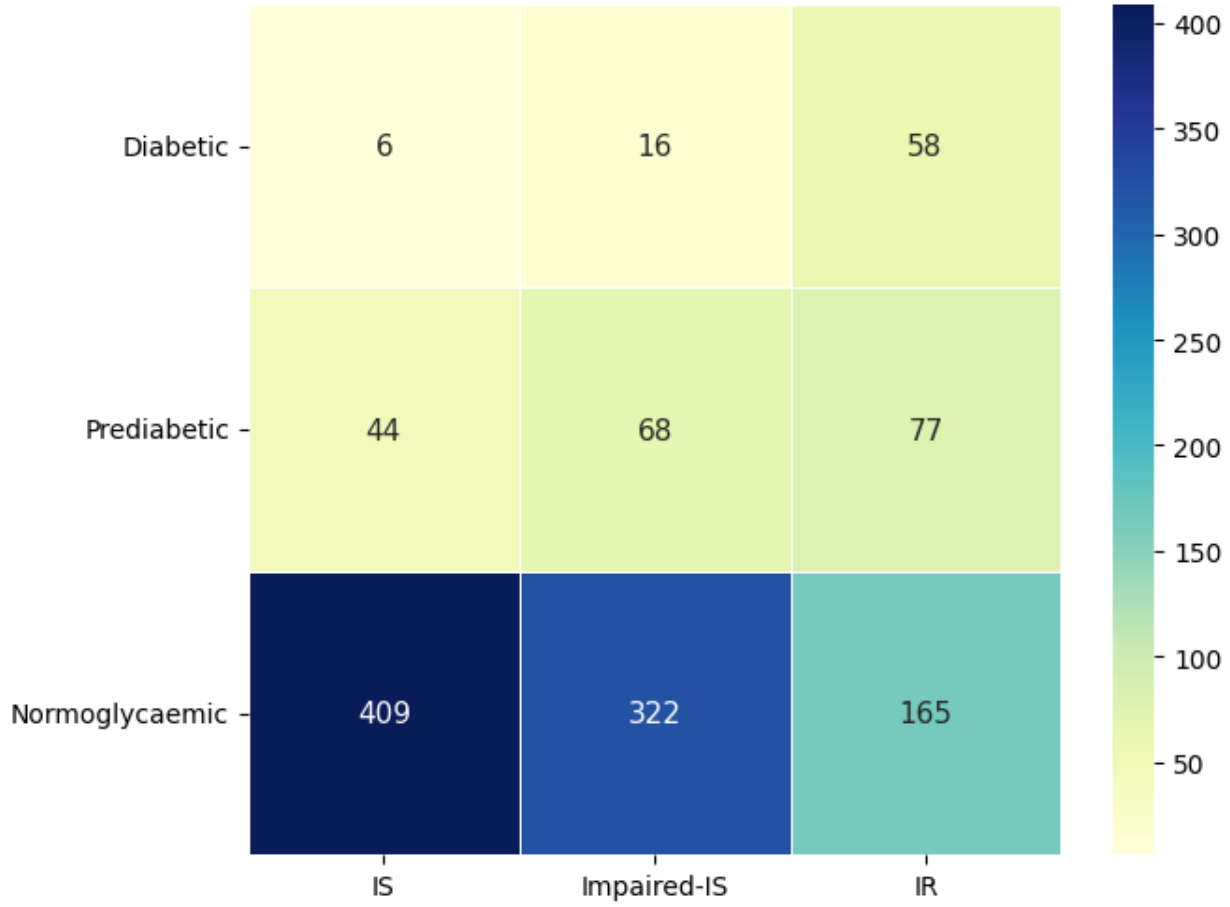
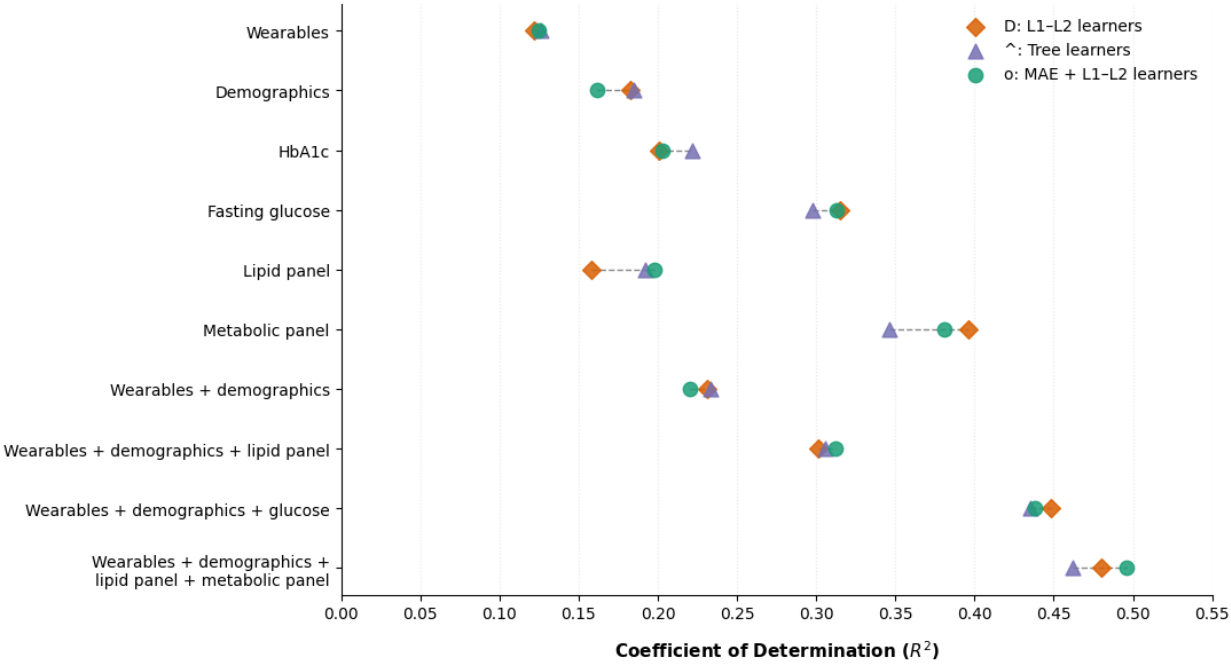
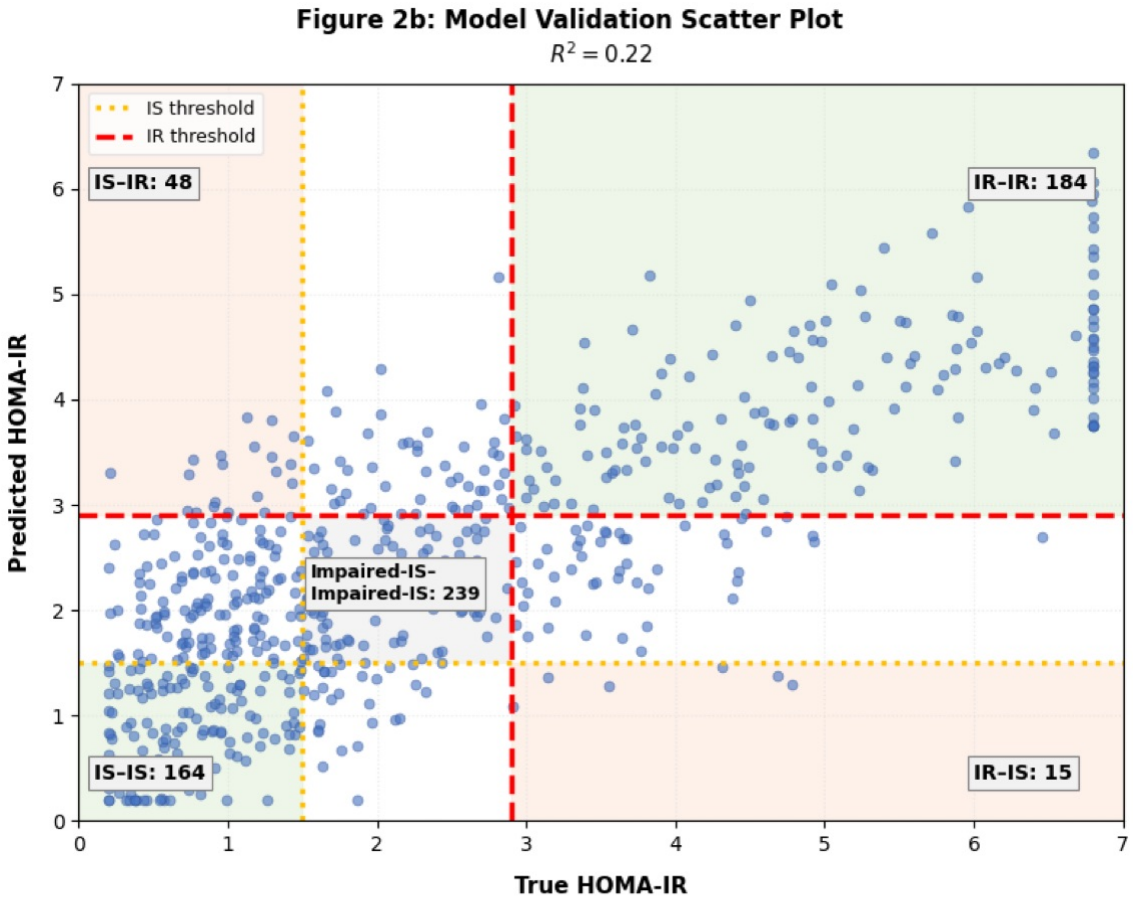
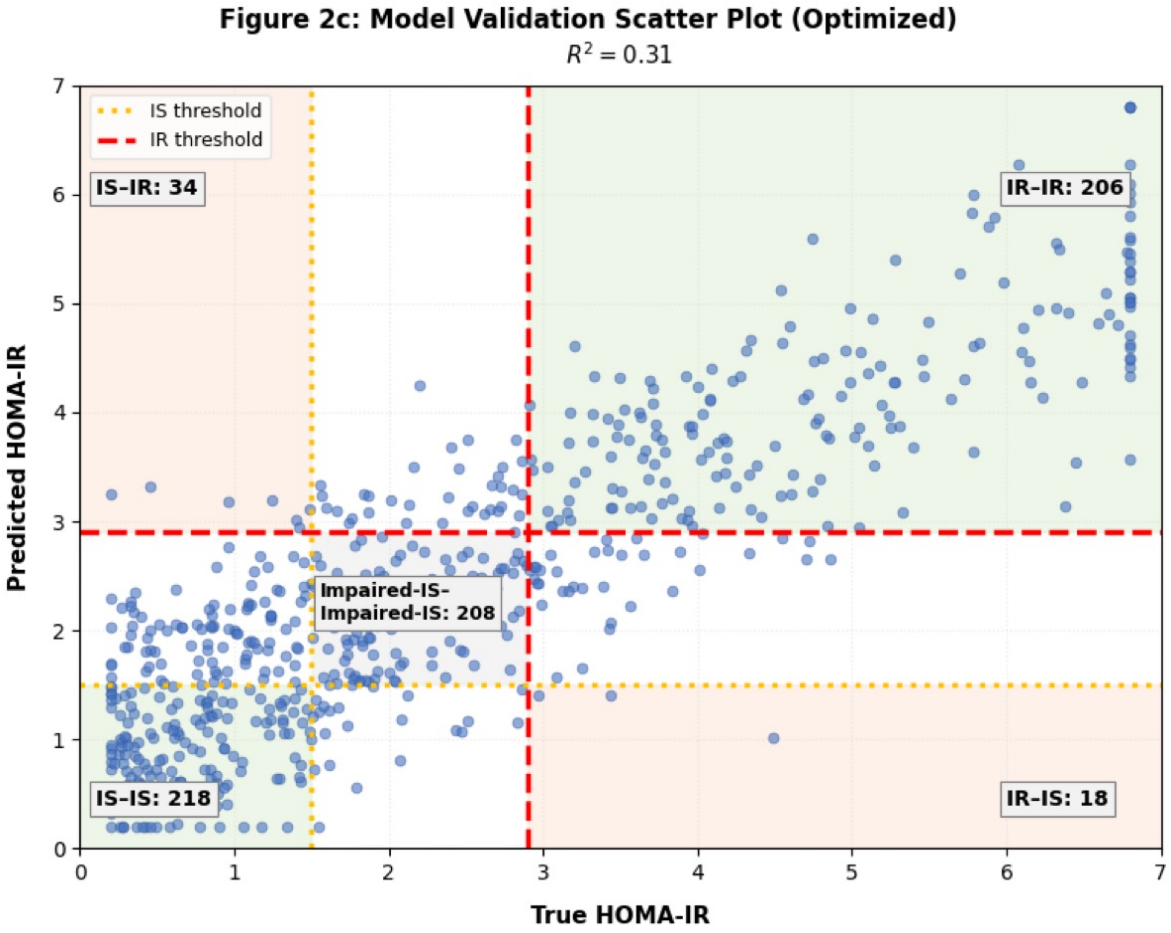


Figure 2a: Model Performance Comparison (R^2)




Figure 2b: Model Validation Scatter Plot
$R^2 = 0.22$
IS threshold
IR threshold
IS–IR: 48
IR–IR: 184
Impaired-IS–
Impaired-IS: 239
IS–IS: 164
IR–IS: 15
Predicted HOMA-IR
True HOMA-IR
0
1
2
3
4
5
6
7


Figure 2c: Model Validation Scatter Plot (Optimized)
$R^2 = 0.31$
IS threshold
IR threshold
IS–IR: 34
IR–IR: 206
Impaired-IS–
Impaired-IS: 208
IS–IS: 218
IR–IS: 18
Predicted HOMA-IR
True HOMA-IR
0
1
2
3
4
5
6
7

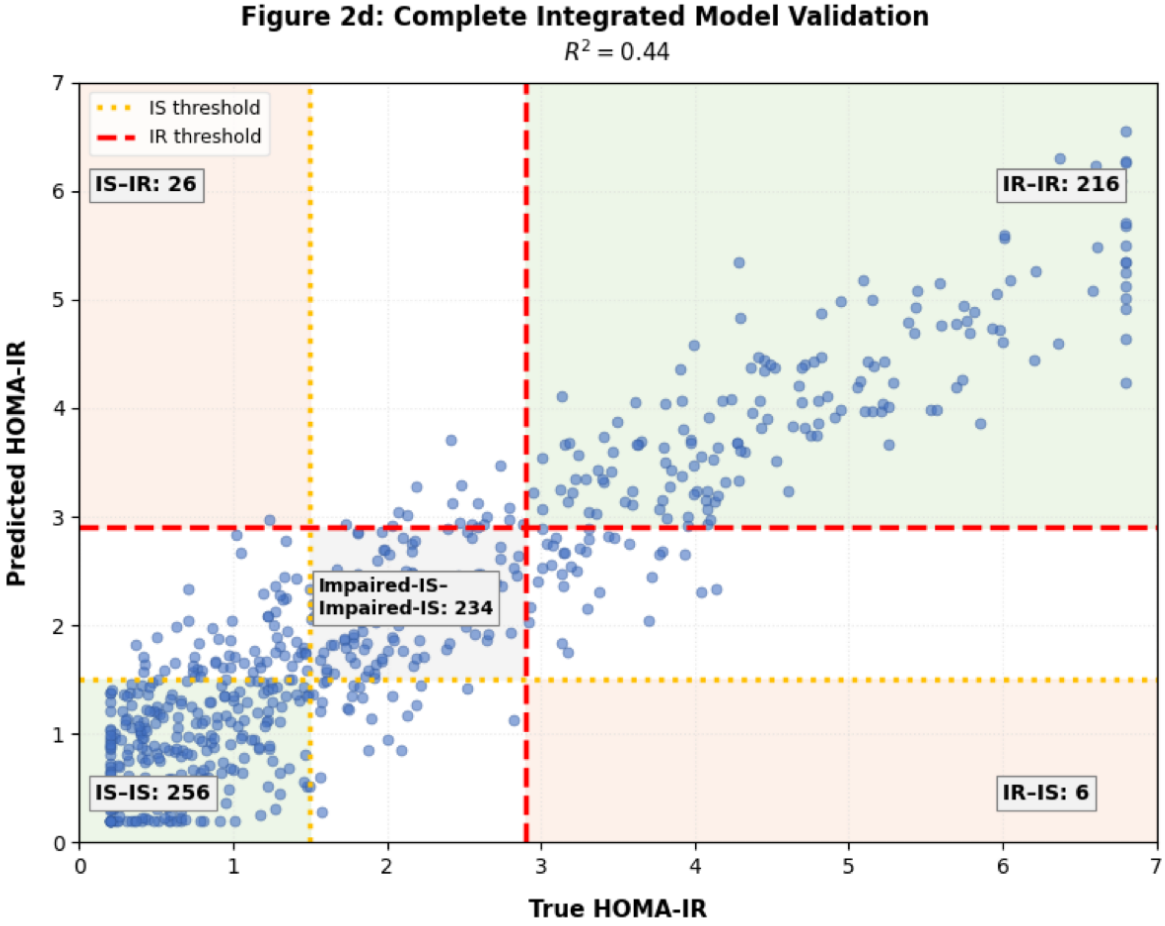

Figure 2d: Complete Integrated Model Validation
$R^2 = 0.44$
IS threshold
IR threshold
IS–IR: 26
IR–IR: 216
Impaired-IS–
Impaired-IS: 234
IS–IS: 256
IR–IS: 6
Predicted HOMA-IR
True HOMA-IR
0
1
2
3
4
5
6
7

Figure 2e: Ultimate Optimized Model Validation

$R^2 = 0.50$

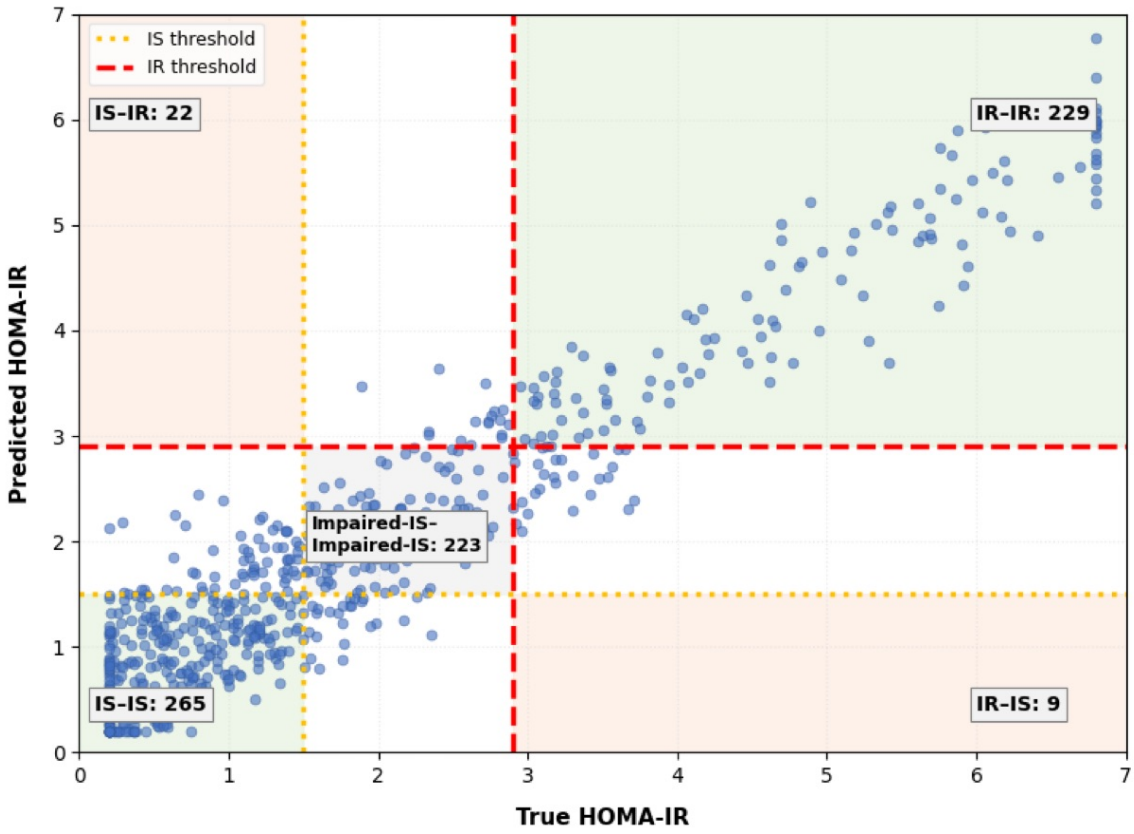
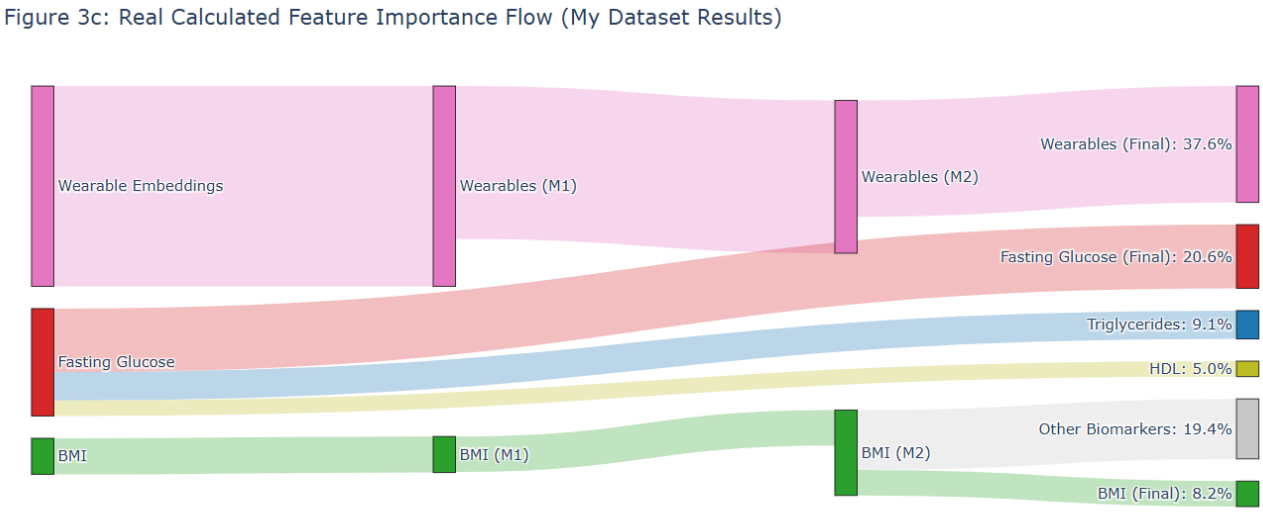


Fig 3c code:


Figure 3c: Real Calculated Feature Importance Flow (My Dataset Results)
Wearable Embeddings
Wearables (M1)
Wearables (M2)
Wearables (Final): 37.6%
Fasting Glucose
Fasting Glucose (Final): 20.6%
Triglycerides: 9.1%
HDL: 5.0%
BMI
BMI (M1)
BMI (M2)
Other Biomarkers: 19.4%
BMI (Final): 8.2%

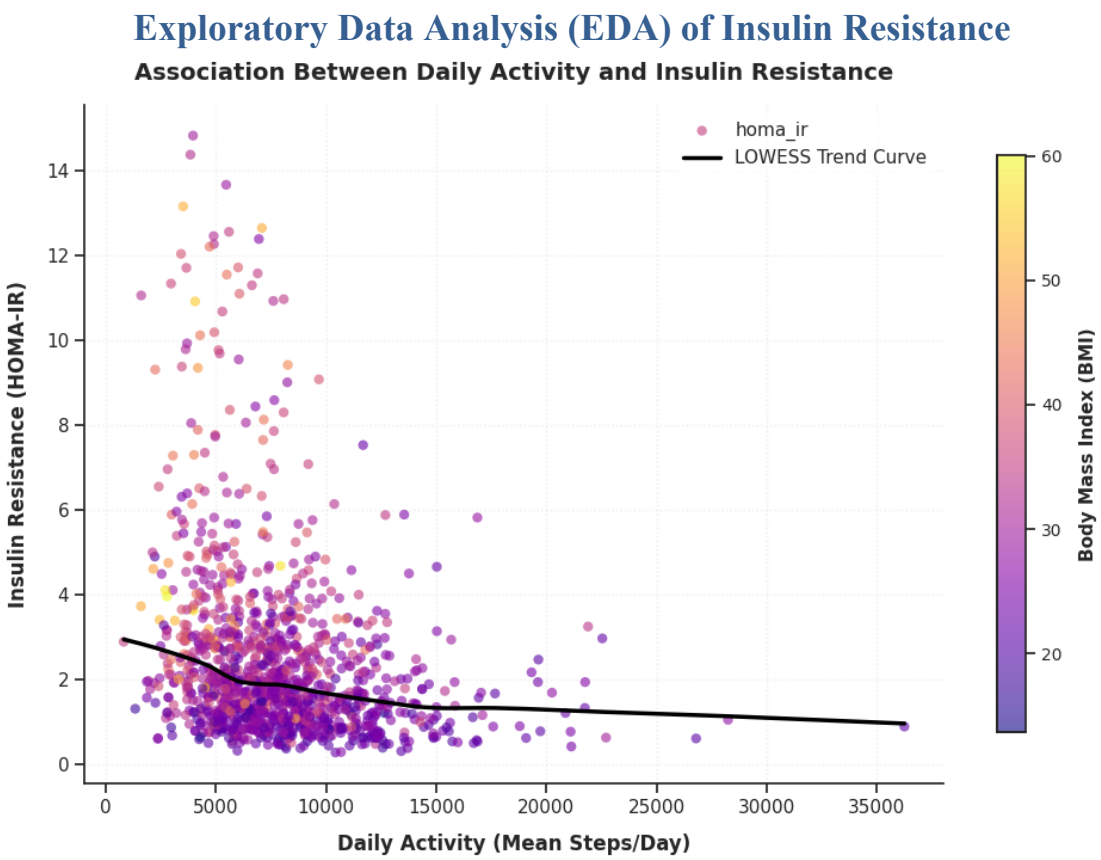


Figure 1. Association Between Daily Activity and Insulin Resistance. The scatter plot illustrates the relationship between mean daily step count and HOMA-IR values. The LOWESS trend line suggests an overall inverse association, indicating that higher levels of habitual physical activity are generally associated with lower insulin resistance. Point coloration represents BMI, highlighting the potential confounding influence of adiposity on this relationship.

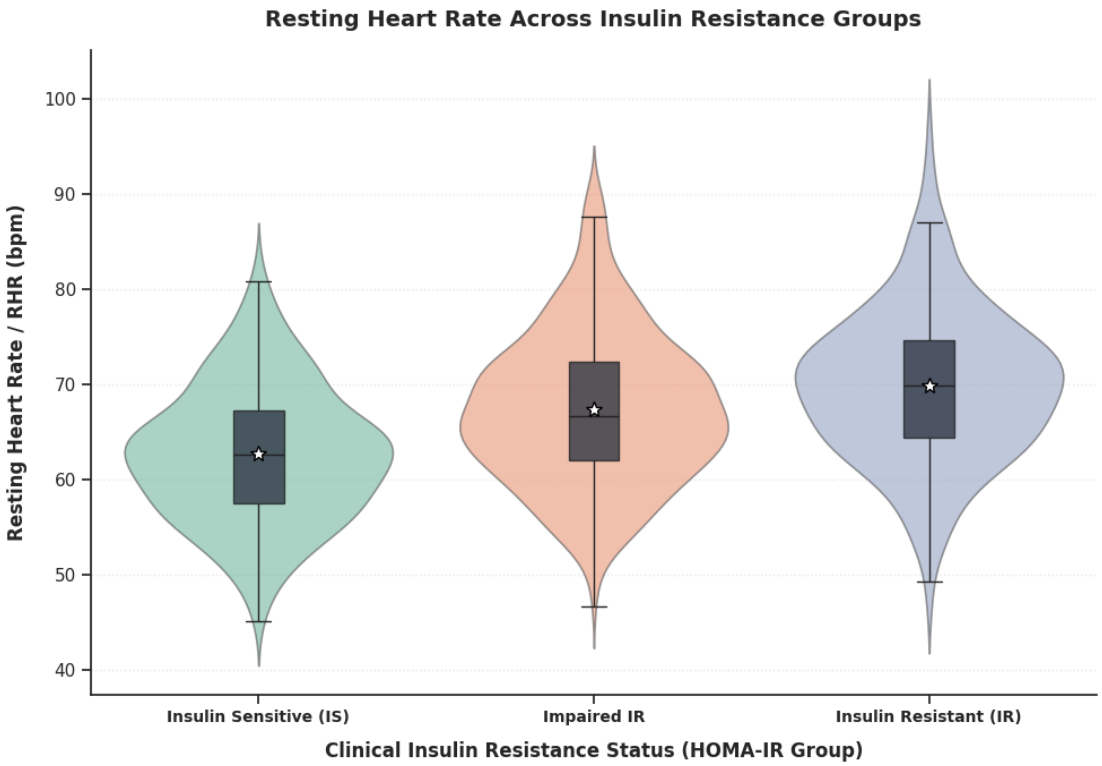


Figure 2. Resting Heart Rate Across Insulin Resistance Groups. Violin plots demonstrate the distribution of resting heart rate (RHR) across clinically defined insulin sensitivity categories. Individuals classified as insulin resistant exhibit higher central tendency and broader variability in RHR compared with insulin-sensitive participants, suggesting a potential link between autonomic dysfunction and metabolic impairment.

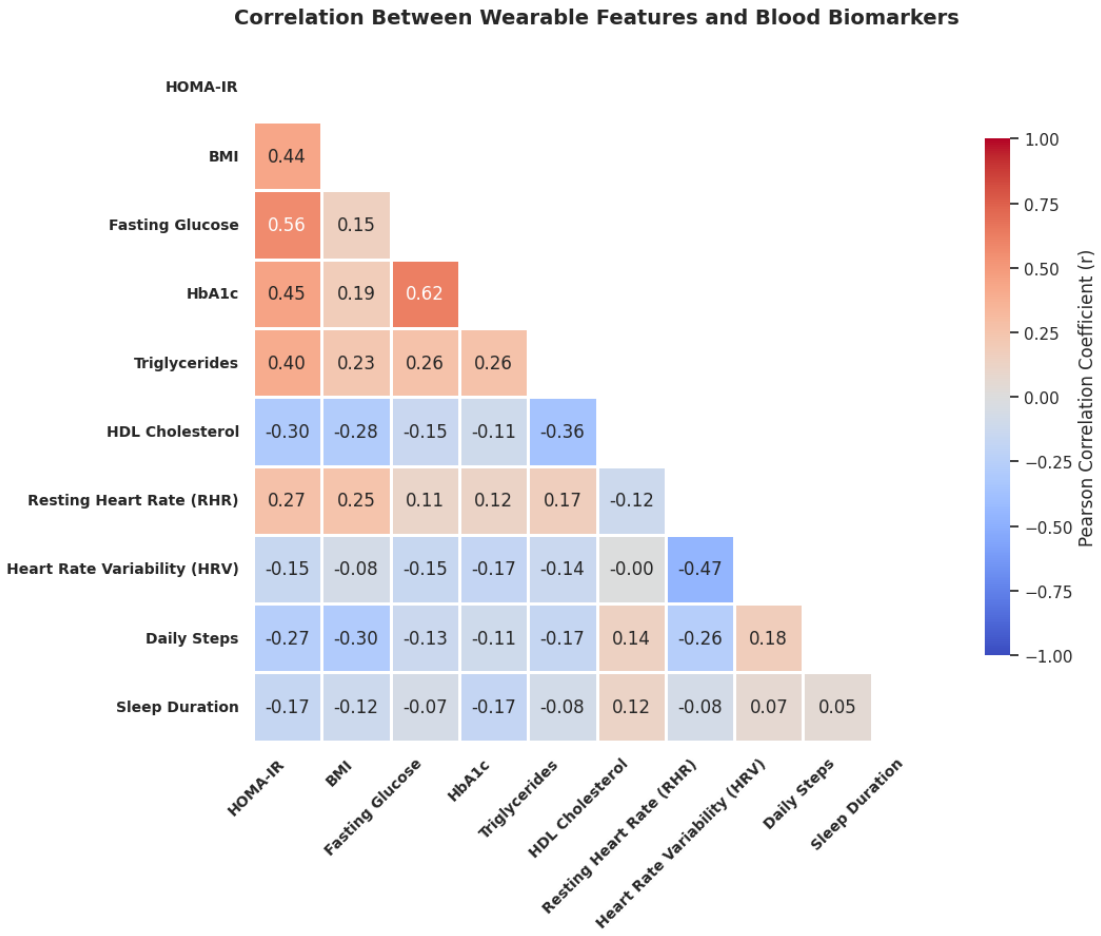


Figure 3. Correlation Matrix of Wearable Features and Blood Biomarkers. Pearson correlation coefficients reveal moderate positive associations between HOMA-IR and traditional metabolic risk markers, including fasting glucose, HbA1c, BMI, and triglycerides. Conversely, HDL cholesterol, daily step count, and heart rate variability display negative correlations with insulin resistance, supporting their potential protective roles.

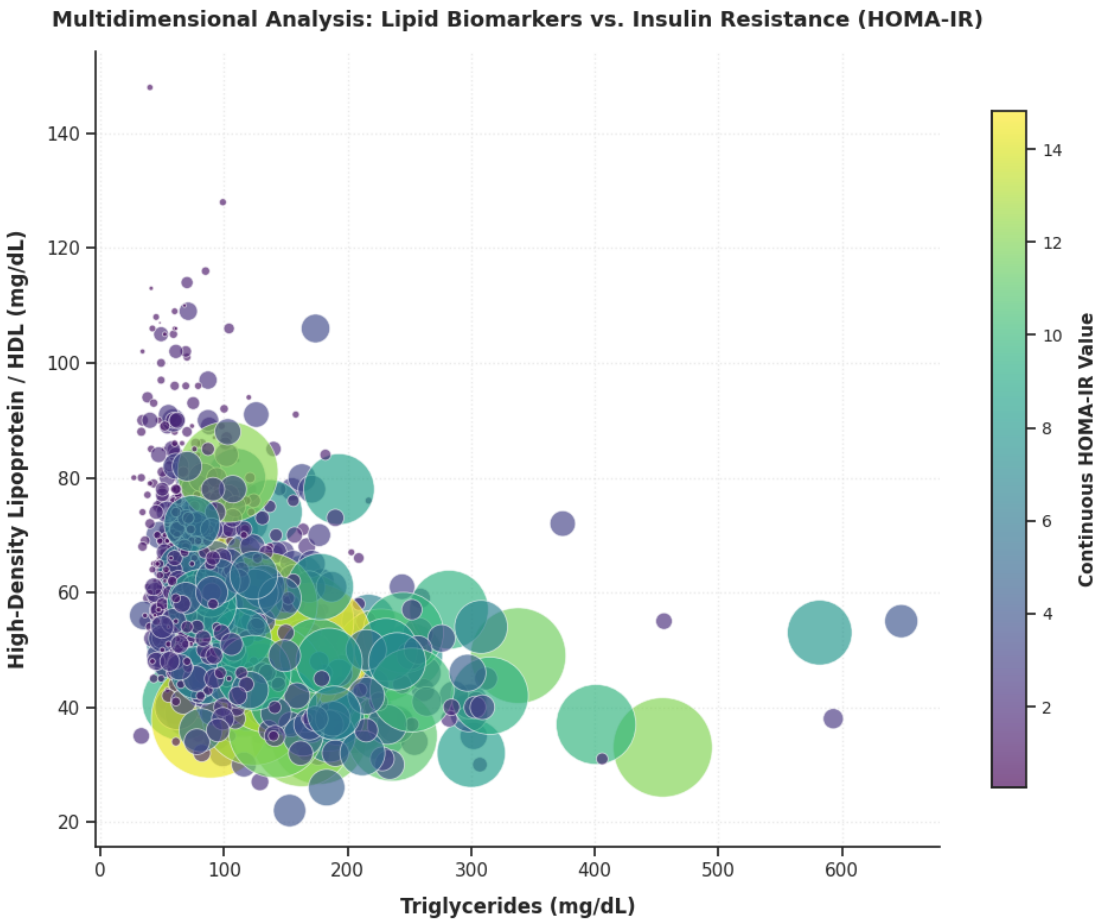


Figure 4. Multidimensional Relationship Between Lipid Biomarkers and Insulin Resistance. Bubble size and color intensity correspond to continuous HOMA-IR values. Participants with elevated triglyceride concentrations and reduced HDL cholesterol levels generally demonstrate higher insulin resistance, reflecting the dyslipidemic phenotype commonly observed in metabolic dysfunction.

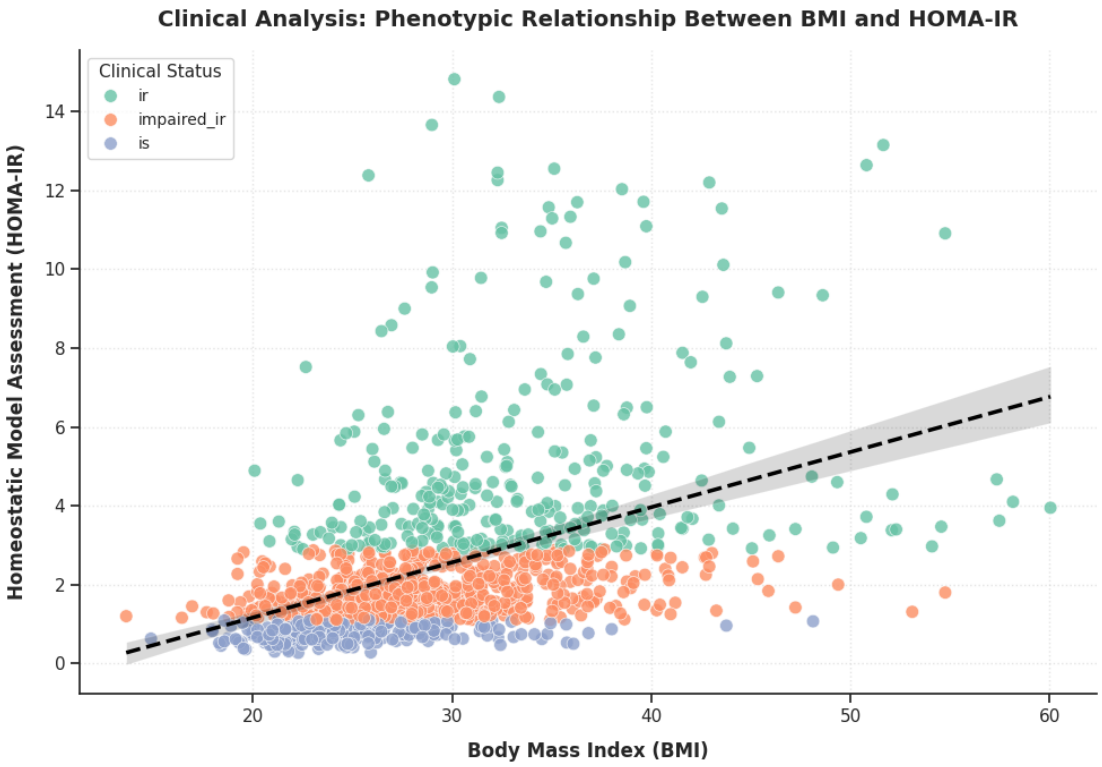


Figure 5. Phenotypic Relationship Between Body Mass Index and HOMA-IR. The regression analysis indicates a positive association between BMI and insulin resistance. Higher BMI values are associated with progressively elevated HOMA-IR measurements, while clinical group stratification highlights the increasing prevalence of insulin resistance among individuals with greater adiposity.

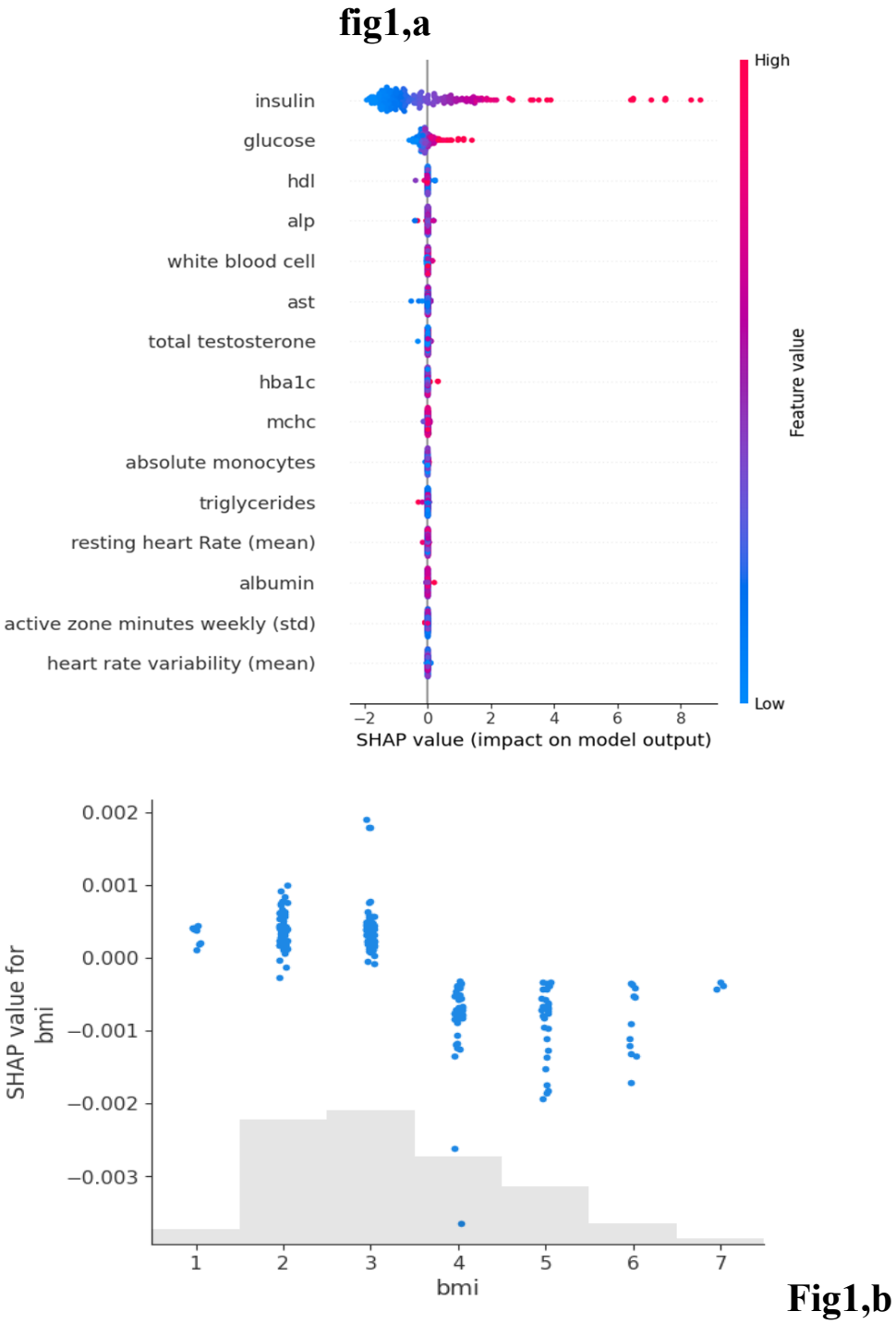


Figure 1: SHAP explanation of feature importance for HOMA-IR prediction. (A) SHAP summary plot illustrating the feature importance and the direction of impact for the top 15 predictors in the HOMA-IR regression model. Each point represents an individual participant, where the color denotes the feature value (red for high, blue for low). High levels of routine clinical biomarkers, specifically **insulin** and **hba1c**, followed by physical metrics such as **bmi** and **age**, exhibit the strongest positive SHAP values, driving the model's prediction toward a higher HOMA-IR score. Conversely, physiological features derived from wearable devices (e.g., sleep duration and heart rate variability metrics) show a distributed but noticeable secondary impact. **(B)** SHAP global feature importance bar plot displaying the mean absolute SHAP values, quantifying the overall magnitude of each feature's contribution to the final insulin resistance estimation.

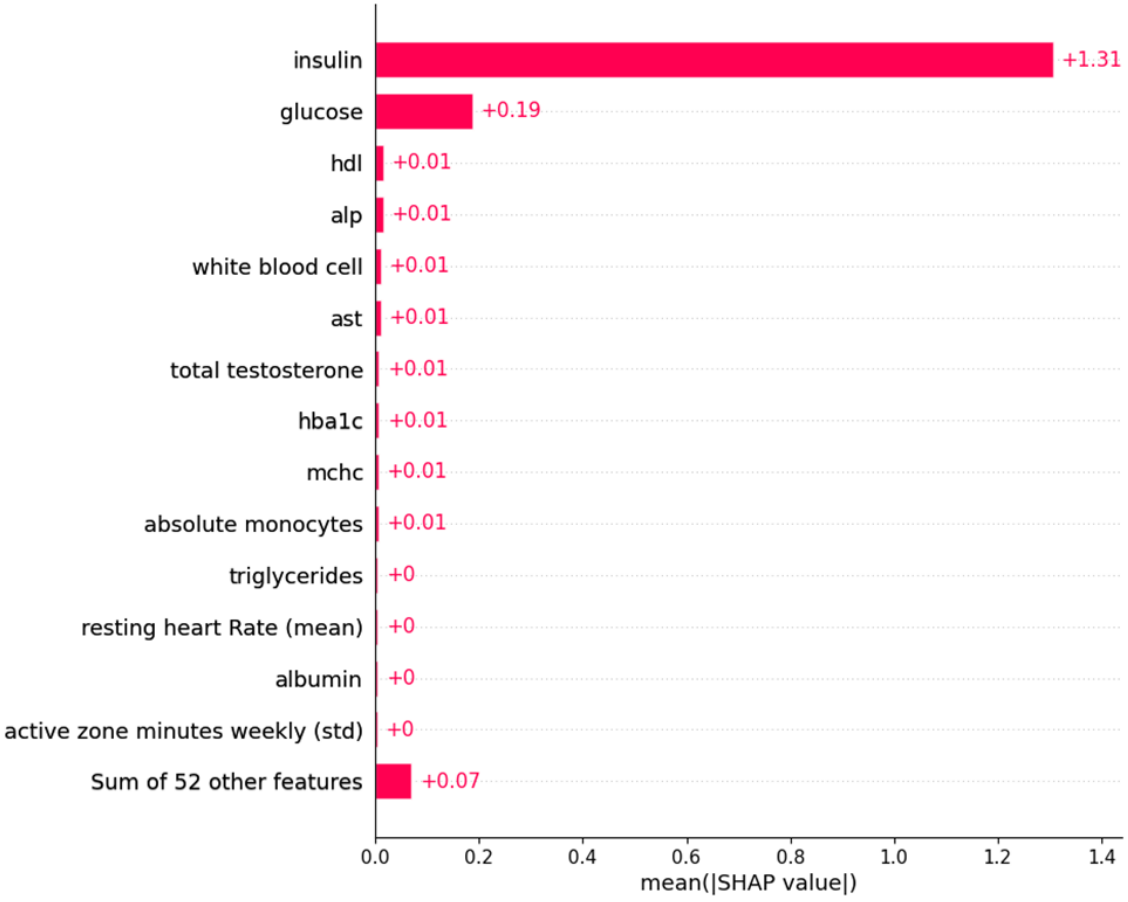


Fig2

Figure 2: SHAP feature importance bar plot displaying the mean absolute SHAP values for the most influential predictors of insulin resistance. This plot quantifies the average magnitude of each feature's contribution to the final HOMA-IR prediction, regardless of the direction of the effect. Clinical blood biomarkers (insulin and hba1c) emerge as the primary drivers of the model's variance, with bmi serving as the most critical anthropometric factor. The chart highlights the complementary role of continuous wearable-derived physiological signals, which provide essential behavioral and baseline metadata to refine the deep learning predictive framework.

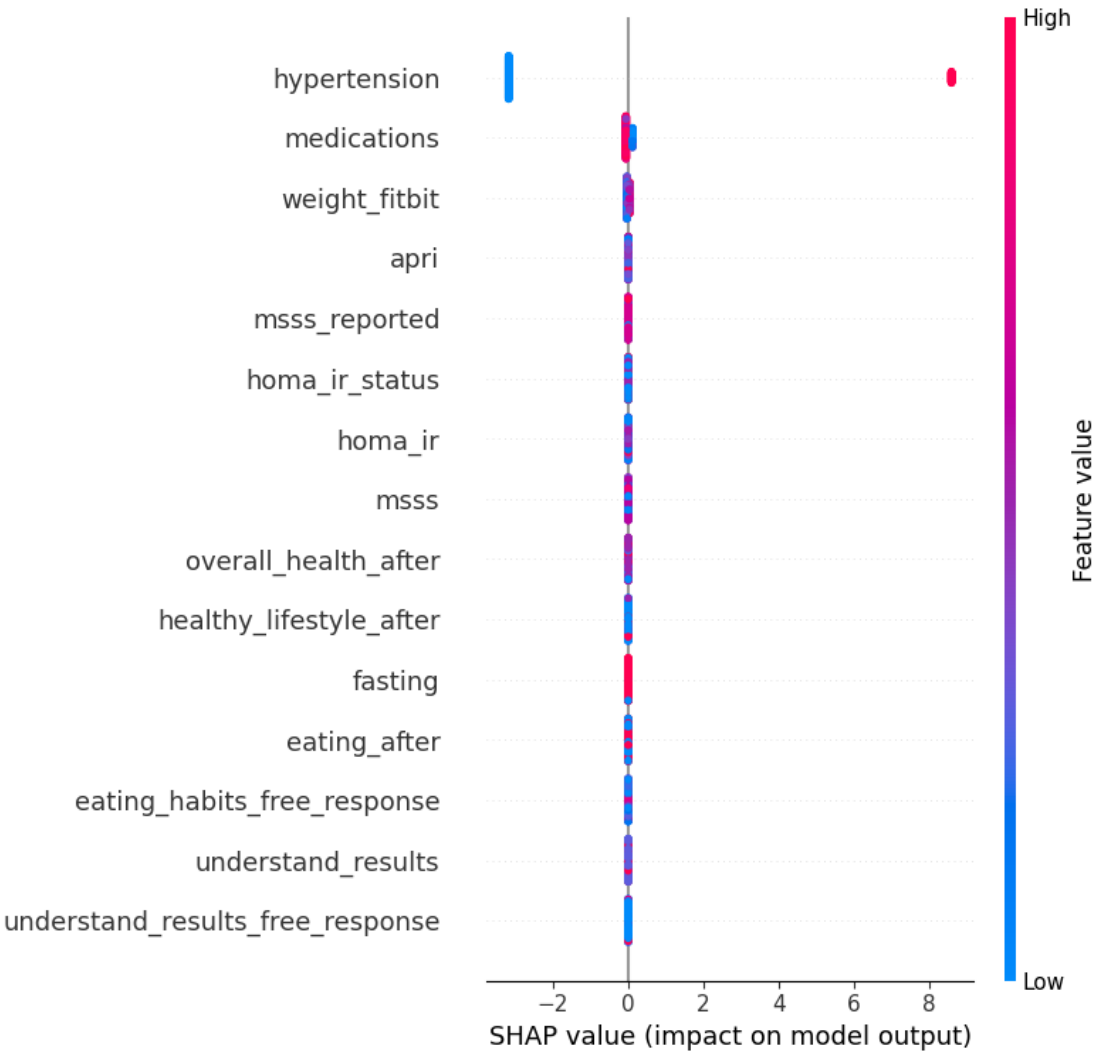


Fig3

Figure 3: SHAP summary plot for the binary classification of hypertension. The features are arranged in descending order of their impact on the model's output log-odds. The risk score framingham30_risk is heavily dominant, where high values (red) sharply increase the probability of a positive hypertension prediction. Demographic factors like age also demonstrate a clear positive correlation with elevated risk. In contrast, stable and lower resting heart rate variations (indicated by specific RHR_FreeLiving metrics) are clustered on the negative SHAP side, signifying a protective or low-risk association with hypertension.

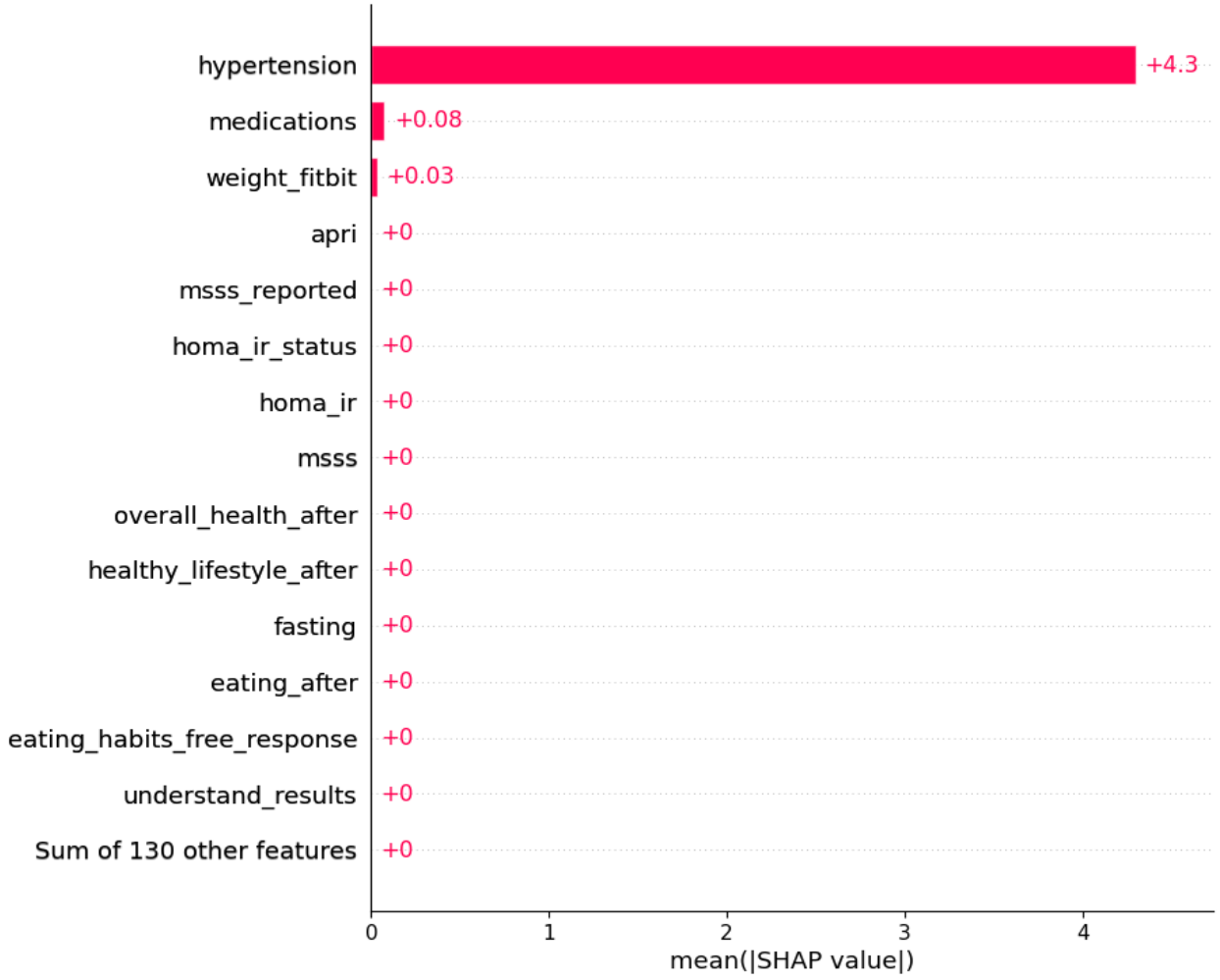


Fig4

Figure 4: SHAP bar plot for the hypertension classification model, depicting the mean absolute SHAP values across the test dataset. The framingham30_risk index exerts a commanding influence over the classification outcomes, substantially outweighing other features with an average absolute impact score exceeding (1.4). age represents the second most significant standalone factor (0.23), while multi-parametric wearable metrics (such as free-living heart rate standard deviation and daily step averages) contribute a steady, incremental baseline to the model's overall diagnostic capacity.

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